

Project Design Phase

Problem – Solution Fit Template

Date	12 july 2026
Team id	
Project Name	Smart Agriculture
Maximum marks	4 marks

Problem–Solution Fit (Smart Agriculture)

Smart Agriculture focuses on solving the major challenges faced by farmers, such as inefficient irrigation, unpredictable weather conditions, pest and disease outbreak. This solution directly meets farmers' needs by improving productivity, reducing resource wastage, lowering farming costs, and enabling informed decision-making, making agriculture more efficient, sustainable, and profitable.

Purpose

- ❑ Solve farmers' challenges by providing AI- and IoT-based solutions tailored to real agricultural conditions.
- ❑ Increase adoption of smart farming technologies through easy-to-use mobile applications and real-time alerts.
- ❑ Improve communication with farmers by delivering timely recommendations, weather updates, and crop management guidance.
- ❑ Enhance crop yield, conserve water, reduce operational costs, and promote sustainable agricultural practices.

Template:

Define CS, & then CC	1. CUSTOMER SEGMENT(S) Who is your customer? For Smart Agriculture, our customers include: • Small and marginal farmers • Medium and large-scale farmers • Agricultural cooperatives • Farm owners and agri-businesses • Agricultural extension officers • Government agriculture departments • Input suppliers (fertilizers, pesticides, seed companies)	CS	6. CUSTOMER CONSTRAINTS What are customer's perceived constraints from taking action or from using our solution? • High cost of smart devices and technology • Lack of technical knowledge and digital literacy • Unreliable internet connectivity in rural areas • Resistance to change from traditional methods • Lack of awareness about smart agriculture benefits • Initial investment and maintenance concerns	CC	5. AVAILABLE SOLUTIONS Which solution(s) are available for the customers when they face the problem? (Existing ways customers solve the problem) • Traditional farming methods • Manual field monitoring • Local agricultural experts and advice • General weather forecast (TV, apps) • Basic soil testing in labs • Chemical-based pest and disease control	AS	Explore AS, Differentiate
	2. JOBS-TO-BE-DONE / PROBLEMS What jobs-to-be-done (or problems) do your customers have? These could be main tasks, urgent needs, or desired outcomes. • Monitor soil health and crop conditions • Manage irrigation efficiently • Detect pests and diseases early • Get accurate weather updates • Increase crop yield and quality • Reduce farming costs and resource wastage	JB	9. PROBLEM ROOT CAUSE What is the root cause (s) of the problem exists? What is the basic story behind the needs or this job? i.e. customers have to do it because of the change in regulation. • Lack of real-time data and insights • Over-dependence on manual processes • Climate variability and unpredictable weather • Limited access to expert knowledge • Poor resource management (water, fertilizers, pesticides) • Inadequate tools for decision-making	RC	7. BEHAVIOUR What does your customer do to address the problem and get the job done? i.e. defining related mind the right state (mindset, occasion, usage and location), including associated customers spend their time on understanding wants (i.e. Overimport). • Regularly visits fields to check crop and soil • Follows advice from local experts • Uses basic mobile apps for weather updates • Applies fertilizers and pesticides based on experience • Irrigates crops on a fixed schedule • Reacts to problems after visible damage occurs	BE	
Identify using TR & EM	3. TRIGGERS What triggers customers to act? i.e. seeing their neighbour reading about pest attack, reading about a more efficient solution in the news. • Sudden drop in crop yield • Pest or disease attack in nearby farms • Unpredictable weather events • Water scarcity • Increase in input costs • Awareness programs, news, social media, success stories	TR	10. YOUR SOLUTION If you are working on an existing business, write down your current solution that solves customers' job-to-be-done. If it's the case, talk to each customer to understand how they see it's relevance and come up with a solution that fits within customer limitations. Write a problem and emotion outcome / behavior.  SMART AGRICULTURE SOLUTION ✓ Real-time monitoring with IoT sensors ✓ AI-based crop disease detection ✓ Automated irrigation system ✓ Weather forecast & alerts ✓ Fertilizer and crop recommendation ✓ Farmer dashboard & mobile app	SL	8. CHANNELS of BEHAVIOUR 8.1 ONLINE What kind of actions do customers take online? (Exact online channels from it) • Mobile apps, websites, social media, online forums • YouTube videos, WhatsApp groups	CH	Extract online & offline CH of BE
	4. EMOTIONS: BEFORE / AFTER How do customers feel when they face a problem or a job and afterwards? i.e. feel, hesitate or confident. • Before: Worried, uncertain, stressed, confused, afraid of loss • After: Confident, informed, relieved, satisfied, empowered, improved livelihood	EM	8.2 OFFLINE What kind of actions do customers take offline? (Exact offline channels from it and use from the customer development). • Field visits, farmer meetings, agri expos • Local agricultural experts, retail stores • Print media, TV, radio programs	CH			